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Short Communication

Effect of air mixing on high-solids anaerobic digestion of cow manure: Performance and mechanism

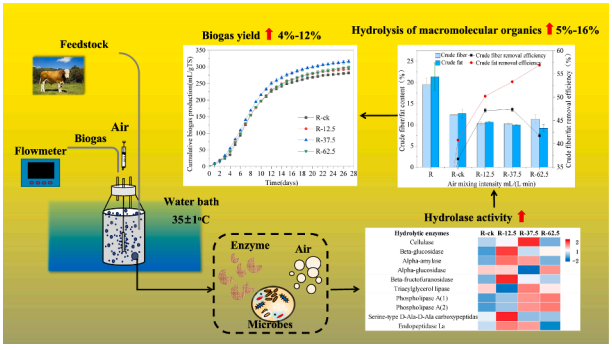
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HIGHLIGHTS

- Effect of air mixing on high-solids anaerobic digestion of cow manure was studied.
- Optimal air mixing intensity of 37.5 mL/(L·min) enhanced CH_4 production by 13 %.
- Air mixing improved the hydrolysis of macromolecular substrate and carbon recovery.
- Air mixing enriched SOB, SAOB and aceticlastic methanogen.
- Predicted hydrolase demonstrated high abundance under air mixing condition.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

This study aimed to further investigate the effect of air mixing on the performance of a high-solids anaerobic digestion system and reveal its underlying mechanisms via analyses of carbon conversion, microbial communities and key functional genes. When the air mixing intensity was 12.5, 37.5 and 62.5 mL/(L·min), compared with the anaerobic digestion without air mixing, the methane yield was increased by 6 %, 13 % and 6 %, respectively. The improved performance was partly attributed to the increased hydrolysis rate of macromolecular substances by 5 %–16 % and carbon recovery in the form of methane by 6 %–7% compared with the controls. Functional flora (*Magnetospirillum*, *Synergistaceae*) and hydrolytic metabolism-related enzymes (cellulose, α-amylase) demonstrated higher abundance under air mixing condition, thus promoting the degradation of organic matter and methane production. This work provides some new insights into the use of air mixing to improve anaerobic digestion of high-solids waste.

1. Introduction

Mixing is an essential part of anaerobic digestion (AD) process and proper mixing can promote mass and heat transfer, improve degradation and transformation efficiency of organic matter and biogas escape (Lindmark et al., 2014). The main AD mixing modes include biogas

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mixing, mechanical mixing and hydraulic mixing (Yang et al., 2021). With regard to long-term operation, biogas mixing is preferred over other mixing methods due to lower energy demand, lower operation cost and higher operation reliability (Dapelo et al., 2015; Yang & Deng, 2020). However, biogas mixing can be affected by its own biogas production, and full mixing cannot be achieved when biogas production is insufficient, which restricts its application in on-farm biogas plants (Yang & Deng, 2020).

In order to solve the problems of insufficient biogas production associated with biogas mixing, Yang & Deng (2020) combined with the latest research progress on the tolerance mechanism of AD system to trace oxygen (this process is generally referred to as "micro-aeration") (Fu et al., 2020; Zhou et al., 2021; Zhu et al., 2022b) put forward a new mixing method, air mixing. Compared with micro-aeration, air mixing can improve mass transfer and create a micro-oxygen environment. Air mixing led to an increase both in methane (CH_4) yield by 6 %–20 % and in hydrogen sulfide (H_2S) removal rate from biogas by 63 %–70 % compared to the other mixing modes (Yang & Deng, 2020). The mass transfer and micro-oxygen environment established by air mixing contributed to increases in CH_4 yield at rates of 63 % and 37 %, respectively (Yang & Deng, 2020). In addition, the stable operation and good biogas production of a full-scale continuous stirred tank reactor using air mixing for more than one year also verified the feasibility of air mixing (Yang et al., 2021).

However, previous studies have mainly applied air mixing or micro-aeration to the low-solids anaerobic digestion system with a total solids (TS) concentration between 2.5 % and 6.0 % (Fu et al., 2016; Zhu et al., 2022a; Zeb et al., 2022). Thus, it was not possible to comprehensively assess the applicability of air mixing in the AD process. In addition, the optimal process parameters of air mixing require further study. If the air mixing intensity is too high, it will have a certain negative impact on anaerobic digestion and reduce the output of biogas (Lindmark et al., 2014). Therefore, the purpose of this study was to investigate the effect of different air mixing intensities on the high-solids AD of cow manure by analyzing and comparing the hydrolysis efficiency and CH_4 production.

2. Materials and methods

2.1. Substrate and inoculum

Fresh cow manure was collected from a beef cattle farm located in Jianyang, China and used as the substrate in this experiment. The cow manure was stored in a refrigerator at 4℃ before use. The TS concentration and volatile solids (VS) concentration of cow manure were 21.8 % and 16.6 %, respectively. The sludge collected from a biogas plant treating animal wastewater was used as inoculum, with TS and VS concentrations of 10.2 % and 5.0 %, respectively (see Supporting Information).

2.2. Experimental design

To conduct the AD batch experiments, 8 glass bottles (each 1.0 L with a working volume of 0.8 L) were used (see Supporting Information). An aeration head was included in the glass bottle for air intake. All glass bottles were inserted into a thermostatic water bath at 35 ± 1 °C. Before AD, 75 g (wet weight) cow manure and 600 g (wet weight) inoculum were mixed in glass bottles. The amount ratio of inoculum to substrate (both expressed in g of VS) was 2:1 and the final TS concentration of the AD system was approximately 10 %. Biogas production with inoculum only (i.e., no cow manure) was also carried out as the blank group.

Four treatments were conducted with air mixing intensity of 0 (i.e., control), 12.5, 37.5 and 62.5 mL/(L·min), and were referred to as R-ck, R-12.5, R-37.5 and R-62.5, respectively, with two in parallel in each group. Air was manually injected into the bottles using a syringe once a day for 1 min each time.

2.3. Analysis method

The daily biogas production was measured with a gas flowmeter (Ritter, Germany), and the amount of injected air for mixing was deducted from daily biogas production. The methods used to determine TS, VS biogas composition, crude fiber and crude fat were according to a previous study (Yang & Deng, 2020).

The analysis of mass balance was performed to determine the carbon (C) recovery after AD of cow manure under different air mixing intensities. For carbon mass balance analysis, the mixed samples before AD (cow manure and sludge) and after AD (digestate and sludge) were collected from the treatment groups and control group. The carbon content (gC) in mixed samples was determined using a total organic carbon (TOC) analyzer (TOC-VCPH, Shimadzu). CH_4 and CO_2 data from biogas production at normal temperature and pressure (273°K and 1 atm) were converted to gC using the ideal gas law equation. The conversion details can be found in a previous study (Nguyen et al., 2019). The carbon mass balance in the treatment groups and control group was analyzed based on the principle of mass conservation.

2.4. Analysis of microorganisms

In order to avoid interference due to residual DNA from dead cells in the AD system on microbial analysis, RNA extracted from sludge samples was selected for high-throughput sequencing in this study. The sludge sample was collected in a centrifuge tube at the end of the experiment. RNA was rapidly extracted according to the E.Z.N.A.® Soil RNA Mini Kit (Omega Bio-Tek, USA). The purified RNA was subsequently used to synthesize complementary DNA (cDNA) using HiScript® Q RT SuperMix for the qPCR kit (Vazyme). The high-throughput sequencing of cDNA samples was performed on the Illumina Miseq platform (Majorbio, China). The raw sequence data files were deposited at NCBI GenBank project (PRJNA906293). The detailed method description and bioinformatics analysis were described in a previous study (Yang & Deng, 2020).

Functional prediction analysis was performed on the Majorbio Cloud Platform (https://www.majorbio.com/). The PICRUSt2 (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States) was applied to evaluate the expression of key enzymes in hydrolytic metabolism based on 16S rRNA data, and the KEGG (Kyoto Encyclopedia of Genes and Genomes) database was used to support functional gene profiling.

3. Results and discussion

3.1. Biogas production performance

Fig. 1a shows the changes in cumulative biogas production following AD of cow manure under different air mixing intensities. The maximum cumulative biogas production (P_{max}) of experimental groups R-12.5, R-37.5 and R-62.5 was 294, 316 and 299 mL/gTS, respectively, and was increased by 4.3 %, 12.1 % and 6.0 % compared with the control group (282 mL/gTS), respectively (see Supporting Information). The maximum cumulative methane production ($P_{max\text{-}CH4}$) of experimental groups R-12.5, R-37.5 and R-62.5 was 185, 198 and 186 CH_4/gTS, respectively. Compared with the control group (175 mL CH_4/gTS), the methanogenic efficiency was increased by 5.7 %, 13.1 % and 6.3 %, respectively (see Supporting Information). Other studies have also reported that micro-aeration improved methane yield from cellulosic substrates (Zhan et al., 2023; Zhu et al., 2022c; Fu et al., 2016). However, direct comparisons between these studies were difficult due to the different compositions of the substrates and operational conditions.

The above results indicated that when the air mixing intensity was 37.5 mL/(L·min), the CH_4 production was markedly improved. Although the CH_4 content in this group was reduced to 53 % due to the dilution of injected air, it did not affect the normal use of biogas (>50 %)

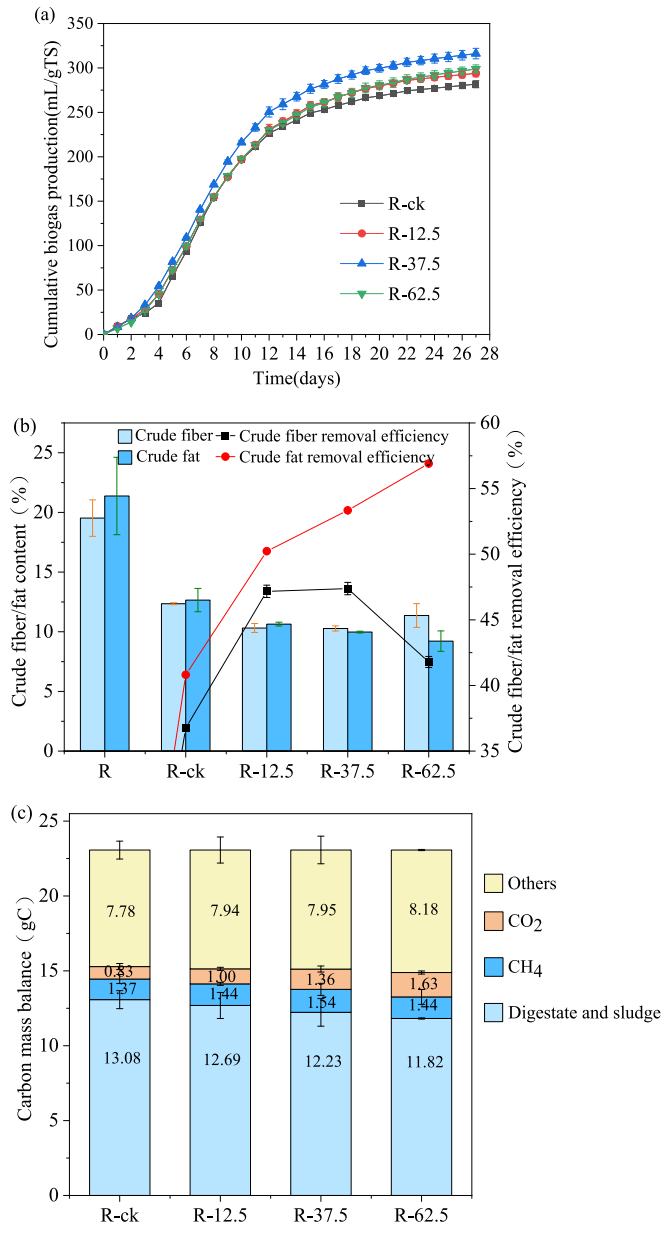


Fig. 1. Variation of cumulative biogas production (a), crude fat/fiber content and crude fat/fiber content removal efficiency (b) and carbon mass balance (gC) (c) under different air mixing intensities. Error bars represent standard deviations of duplicate tests (n = 2). R-ck represents the control group without mixing; R-12.5, R-37.5 and R-62.5 represent the experimental groups with air mixing intensity of 12.5, 37.5 and 62.5 mL/(L·min), respectively.

(Nallamothu et al., 2013). However, when the intensity of air mixing was high, the efficiency of improvement decreased. This may be because the excess O_2 was not timely consumed by facultative bacteria in the early stage of AD, which does not produce a strict anaerobic reducing environment, thus inhibiting the activity of strict anaerobic methanogens. Alternatively, facultative bacteria perform aerobic respiration, converting organic matter to CO_2 instead of CH_4 (Botheju & Bakke, 2011).

3.2. Hydrolysis of macromolecular substances

The hydrolysis of macromolecular substances into soluble organics is generally considered as the rate-limiting step during AD of feedstock with a high-solids concentration (Yu et al., 2021). In this study, samples of cow manure AD on day 0 (marked as R) and day 23 were taken to determine the content of crude fiber and crude fat. The results showed that air mixing promoted the removal efficiency of crude fiber and crude fat (see Fig. 1b). Compared with control group R-ck, the removal rates of crude fiber in experimental groups R-12.5, R-37.5 and R-62.5 were increased by 10 %, 11 % and 5 %, and the removal rates of crude fat were increased by 9 %, 13 % and 16 %, respectively. The removal rate of crude fiber was the highest at the air mixing intensity of 37.5 mL/(L·min), and showed a downward trend at the mixing intensity of 62.5 mL/(L·min). Zeb et al. (2022) also reached a similar conclusion where the lignin removal rate decreased as the micro-aeration dose increased. This may be due to the damaging effect of high oxygen flow on the lignin hydrolysis process, but the specific reasons for this require further investigation.

3.3. Carbon conversion

In the AD process, it is hoped that the carbon from feedstock can be converted into CH_4 as much as possible by microorganisms (Tian et al., 2022). Therefore, a carbon mass balance analysis was conducted in this study, and the results are shown in Fig. 1c. The carbon contents in CH_4 of the R-ck, R-12.5, R-37.5 and R-62.5 groups accounted for 6 %, 6 %, 7 % and 6 % of the total carbon contents in the fermented materials, respectively. These results showed that the biodegradation of lignocellulose and other refractory substances in cow manure was raised under air mixing condition, and more dissolved substrates were converted into biogas, which indicated that air mixing promoted carbon recovery. Moreover, when the air mixing intensity was 37.5 mL/(L·min), the carbon recovered in the form of CH_4 was the highest; when the air mixing intensity was 62.5 mL/(L·min), more carbon was converted to CO_2.

3.4. Microbial community characterization

The relative abundance of bacterial and archaeal communities at the genus level under different air mixing intensities are shown in Fig. 2a and 2b. *Magnetospirillum* is a sulfide-oxidizing bacterium (SOB) (Haosagul et al., 2021). *Smithella*, *Thiovirga*, *Thiokalimicrobium* and *Hydrogenophaga* were also reported to play important roles in sulfide oxidation (Haosagul et al., 2021). The total abundance of these SOBs was 5 %, 5 %, 7 % and 11 % in R-ck, R-12.5, R-37.5 and R-62.5, respectively. According to the KEGG database, sulfide oxidization is likely catalyzed by either sulfide:quinone reductase (EC1.8.5.4) or flavocytochrome c sulfide dehydrogenase (EC1.8.2.3). PICRUSt-based prediction of sulfur metabolism-related enzymes only detected the flavocytochrome c sulfide dehydrogenase in this study. The predicted abundance of flavocytochrome c sulfide dehydrogenase was 45 % to 195 % higher in R-37.5 (756) and R-62.5 (1537) than in R-ck (521). These results showed that air mixing increased the abundance of SOB and related enzyme activities, which was conducive to the removal of H_2S. *Synergistaceae* is a syntrophic acetate-oxidizing bacteria (SAOB), which could oxidize acetic acid to CO_2 and H_2 as substrates for methanogenesis (Zhu et al., 2022a). The abundance of *Synergistaceae* in all groups was 4 % (R-37.5) > 2 % (R-62.5) > 1 % (R-12.5) > 1 % (R-ck). An enrichment of SOB and SAOB is conducive to desulfurization, VFA degradation and methanogenesis.

Methanosaeta was the dominant archaea in all experimental groups, with an abundance of 52 %–61 % (Fig. 2b). Compared with the control group (R-ck), the relative abundance of *Methanosaeta* in the air mixing groups (R-12.5, R-37.5 and R-62.5) increased by 5 %, 16 % and 8 %, respectively, indicating that air mixing favored the growth of *Methanosaeta*. Previous studies on microaerobic systems also demonstrated that *Methanosaeta* was enriched at low oxygen concentrations (Guo et al., 2021). *Methanosaeta* belongs to the acetoclastic methanogens and dominates at low acetate concentrations (Zhou et al., 2021). The oxidation of acetate through aerobic respiration to maintain a dynamic

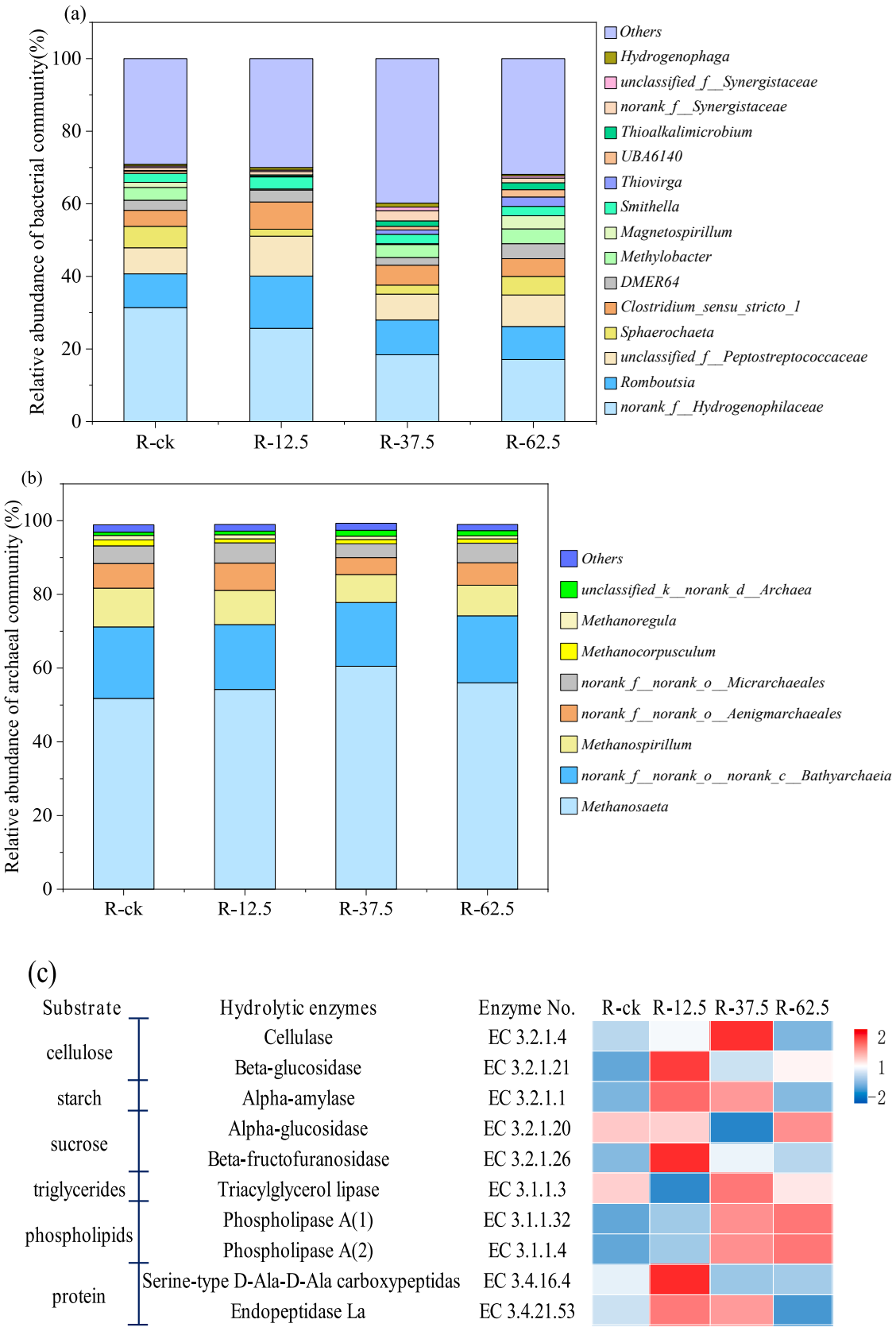


Fig. 2. (a) The relative abundance of bacteria at the genus level, (b) the relative abundance of archaea at the genus level, and (c) schematic diagram of predicted hydrolytic enzymes related to degradation of macromolecules (abundance of each enzyme calculated by normal standardization in the different reactors is represented in a heat map) under different air mixing intensities. R-ck represents the control group without mixing; R-12.5, R-37.5 and R-62.5 represent the experimental groups with air mixing intensity of 12.5, 37.5 and 62.5 mL/(L·min), respectively.

equilibrium of acetate concentration in the presence of a small amount of oxygen, might explain the higher relative abundance of *Methanosaeta* and little accumulation of acetate under air mixing conditions (see Supporting Information).

3.5. Hydrolase prediction

The abundance of functional enzymes related to hydrolytic metabolism was studied by PICRUSt2 based on 16S rRNA gene data in this section. Cellulose and hemicellulose are the main macromolecular organic matter in cow manure and account for approximately 50 % of the total solids (Zhao et al., 2018). Cellulase and beta-glucosidase (EC3.2.1.4 and EC3.2.1.21) have been reported to degrade lignocellulose (Zhu et al., 2022b). Fig. 2c shows that the abundance of cellulase in the R-37.5 was the highest, and was 1.4 times than that in the R-ck, However, the abundance of cellulase in the R-62.5 was the lowest, which might be an important reason for the reduction in the crude fiber removal rate when the air mixing intensity increased to 62.5 mL/(L•min) (see Section 3.2). The predicted beta-glucosidase abundance was 6 % to 19 % higher in the experimental groups (R-12.5, R-37.5 and R-62.5) than in the control group (R-ck) (Fig. 2c). Triacylglycerol lipase (EC3.1.1.3), phospholipase A(1) (EC3.1.1.32) and phospholipase A(2) (EC3.1.1.4) are important lipid hydrolases (Reihaneh et al., 2022). The predicted abundance of lipid hydrolases in this study was very low (see Supporting Information), which might be related to the lower lipid content in cow manure. It was observed that the relative abundance of lipid hydrolase in all groups was R-62.5 > R-37.5 > R-12.5 > R-ck, which was consistent with the change in crude fat removal rate (Fig. 1b). The above results proved that the relative abundance of hydrolysis-related extracellular enzymes in the reactor with air mixing was generally higher than that in the reactor without air mixing, indicating that air mixing stimulated the secretion of extracellular enzymes and accelerated the degradation of macromolecular organic matter and further facilitated the production of CH_4.

4. Conclusion

The effect of air mixing on the high-solids AD of cow manure was successfully investigated. Air mixing increased methane production by 6–13 %, enhanced hydrolysis of macromolecular substrates, and improved carbon recovery. Microbial community structure analysis showed that air mixing optimized microbial community structure, resulted in higher relative abundance of sulfide-oxidizing bacteria (*Magnetospirillum*), syntrophic acetate-oxidizing bacteria (*Synergistaceae*) and aceticlastic methanogen (*Methanosaeta*). Hydrolytic metabolism related enzymes including cellulase, α-amylase, and triacylglycerol lipase predicted by PICRUSt2 based on 16S rRNA gene data demonstrated high abundance of hydrolase under air mixing conditions.

CRediT authorship contribution statement

Xinwei Li: Methodology, Writing – original draft. **Liangwei Deng:** Supervision, Data curation, Funding acquisition, Writing – review & editing. **Fucheng Li:** Funding acquisition, Writing – review & editing. **Dan Zheng:** Funding acquisition. **Hongnan Yang:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2022.128545.

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